EEE 304, LAB 4

The objective of this sequence of laboratory exercises is to explore a few fundamental problems arising in Amplitude Modulation and Communication Systems.

1. Amplitude Modulation

Modulation is the process of varying one or more properties of a periodic waveform, called the carrier signal (typically of very high frequency), with a modulating signal that generally contains information to be transmitted. There are two motivating reasons for modulation:

1. Modulation allows for the use of small antennae in message transmission therefore making the application portable e.g. mobile phones.
2. It also allows us to multiplex, or share, a communication medium among many concurrently active users through the choice of different carrier frequencies separated by a frequency gap band. This technique is known as Frequency Division Multiplexing.

Radio, television, GPS, mobile phones and all other wireless communications devices transmit information across distances using electro-magnetic waves. To send these waves across long distances in free space, the frequency of the transmitted signal must be quite high compared to the frequency of the information signal. This keeps aliasing at bay as well as help keep antenna sizes small. For example, the signal in a cell phone is a voice signal with a bandwidth of about 4 KHz. The typical frequency of the transmitted and received signal is several hundreds of megahertz to a few gigahertz.

Let us look at how the antenna size can be made smaller with higher carrier frequencies. For example, the wavelength of a 1 GHz electromagnetic wave in free space is 30 cm, whereas a 1 kHz electromagnetic wave is one million times larger, 300 km, it would be impossible to build and power such a behemoth!

Communication that uses modulation to shift the frequency spectrum of a signal is known as carrier communication. In this mode, one of the basic parameters (amplitude, frequency, or phase) of a sinusoidal carrier of high frequency w_c is varied in proportion to baseband signal m(t). In the remainder of this lab exercise we will interchangeably use the notations (w and f) to represent frequency. *f* denotes the frequency of a sinusoidal signal in Hz, whereas w is the frequency in rad/sec.

Amplitude modulation is characterized by the fact that the amplitude *A* of the carrier signal, $c(t) = Acos(w_c t + \theta_c)$, is varied in proportion to the amplitude of the modulating (message) signal *m(t)*. The frequency and the phase of the carrier are fixed. For simplicity, we can assume that $\theta_c = 0$ in the carrier signal. If the carrier amplitude A is made directly proportional to the modulating signal *m(t),* we obtain the signal $m(t)cos(w_c t)$. The carrier signal $Acos(w_c t)$ is added to this signal, and the resulting signal

$$\Phi_{AM}(t) = (A + m(t))cos(w_c t)$$

is referred as the amplitude modulated signal. Figure 1 illustrates the amplitude modulation.

The modulating (message) signal is multiplied by the carrier signal. The modulating signal forms the envelope of the modulated signal.

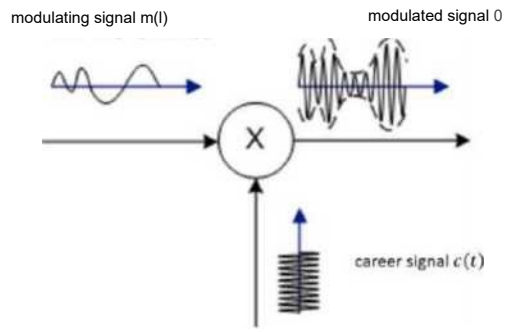


Figure 1. Illustration of Amplitude Modulation.

Let the Fourier transform of the message signal be denoted by M(W), i.e.,

$$F\{m(t)\} = M(jw)$$

From the properties of the Fourier Transform, it follows that

$$F\{m(t)\cos(w_c t)\} = [M(jw + jw_c) + M(jw - jw_c)]/2$$

Hence, the modulated signal transform is given by

$$F\{\Phi_{AM}(t)\} = \frac{[M(jw + jw_c) + M(jw - jw_c)]}{2} + \pi A[\delta(w + w_c) + \delta(w - w_c)]$$

The process of modulation in time and frequency domains is described in Figure 2 and Figure 3. There is a replication of the spectrum of the modulating signal, each centered around $-w_c$ and $+w_c$. A portion that lies above the w_c is called Upper Side Band (USB) and a portion lies below w_c is called Lower Side Band (LSB). For each replication the amplitude is reduced from $2A$ to A.

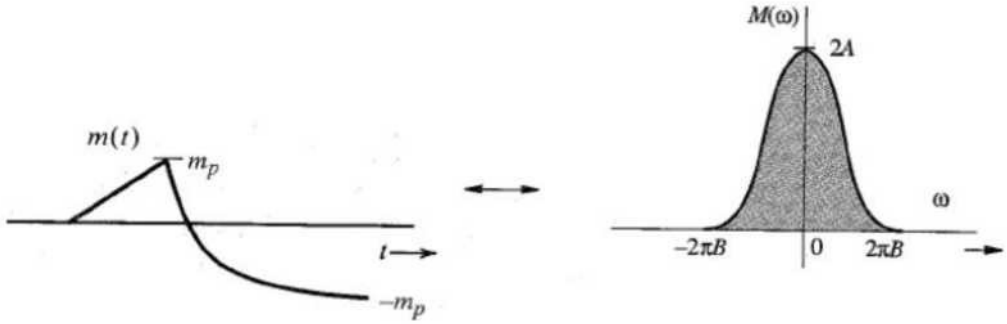


Figure 2. Modulating signal m(t) in time (left) and frequency (right).

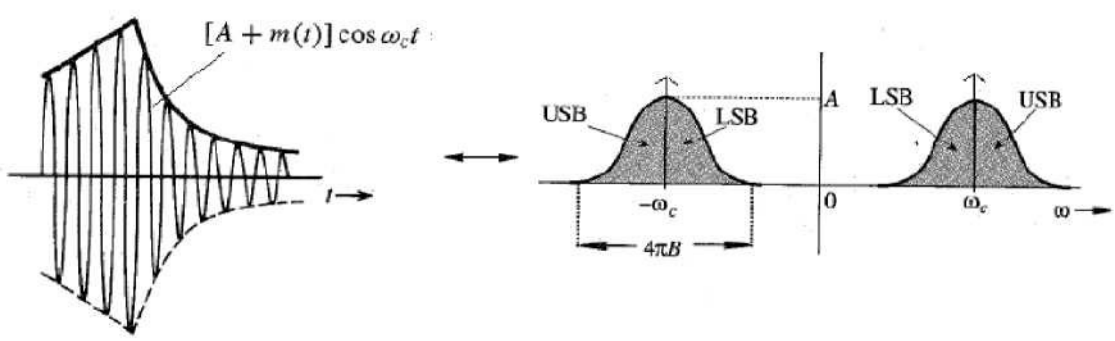


Figure 3. Modulated signal in time (left) and frequency (right).

2. Amplitude Demodulation

Demodulation can be performed in two different ways.

- Coherent Demodulation: This assumes that the carrier signal with same frequency and phase can be generated at the receiver for demodulation.
- Non-Coherent Demodulation: This directly demodulates the received signal from its envelope
  without requiring the carrier signal.

Here, we describe coherent demodulation, as it is very similar to the modulation process and easy to demonstrate and understand. The scheme of modulation shifts the spectrum of the message signal by multiplying it with the carrier signal. Hence, demodulation requires the shifting of the spectrum back to the original location. To achieve this, we multiply again the modulated signal with the carrier signal, as described in Figure 4 and Figure 5.

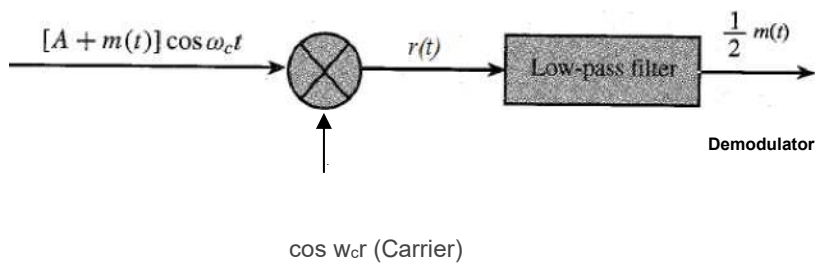


Figure 4. Demodulation.

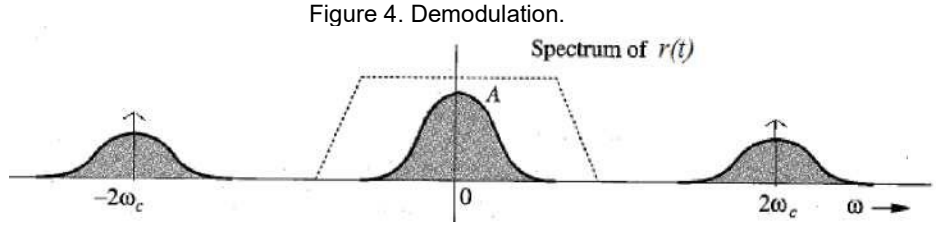


Figure 5. Frequency spectrum of modulated signal multiplied by the carrier. The dashed line indicates the frequency response of a low pass filter used to extract the demodulated signal.

The received signal is

$$\Phi_{AM}(t) = (A + m(t))cos(w_c t)$$

Multiplying with $\cos(w_c t)$ once more, we get the signal r(t) with Fourier Transform

$$R(jw) = \frac{A}{2}\delta(w) + \frac{\pi A}{2}[\delta(w + 2w_c) + \delta(w - 2w_c)] + \frac{1}{2}M(jw) + \frac{1}{4}[M(jw + j2w_c) + M(jw - j2w_c)]$$

It is evident from the last equation that one part of the demodulated signal spectrum is centered at zero frequency and the other at $2w_c$. Hence, the recovery of the original signal $M(jw)$ can be achieved by passing the signal r(t) (with spectrum R(jw)) through a lowpass filter with cutoff frequency greater that the bandwidth “B” of the message signal.

3. Modulation Index

The modulation index (μ) is defined as the ratio between the amplitude of the message signal and the amplitude of the carrier signal. This indicates how much the modulated signal varies around its original level. The value of μ < 1 results in under-modulation and μ > 1 results in over-modulation. Over-modulation results in erroneous signal reconstruction if non-coherent demodulation (envelope detection) is used. In the case of coherent demodulation, any value of μ provides reconstruction.

The following figure shows modulated signals with different modulation index value. The carrier signal has amplitude of 1. The baseband signal has amplitude of 0.5, 1 and 1.5. All baseband signals are shifted up by 1 (amplitude of the carrier signal)

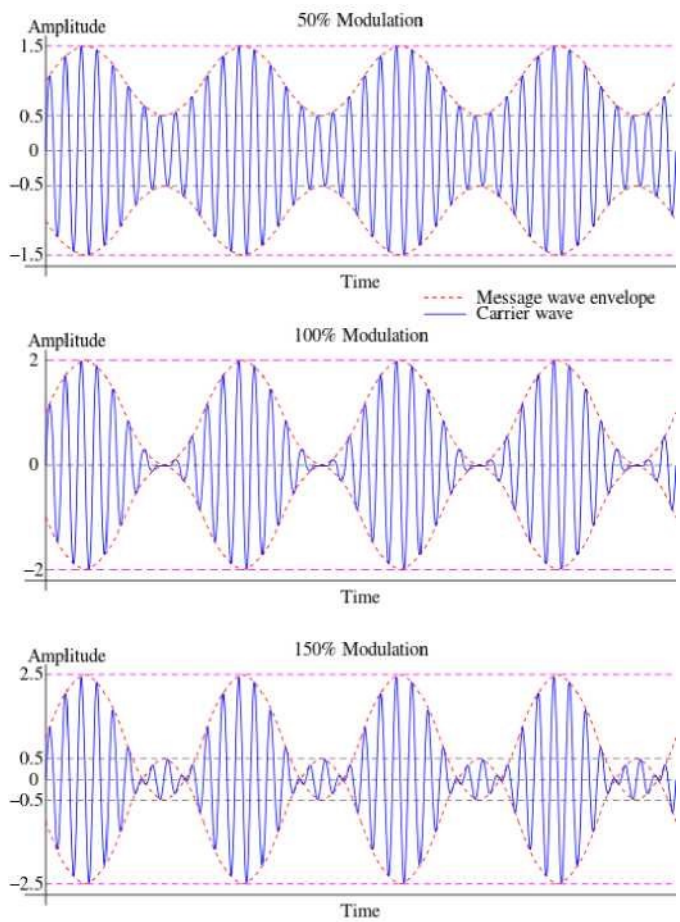


Figure 6. Modulated signal with different modulation index[1]

Reference (other than the textbook)
https://en.wikipedia.org/wiki/Amplitude_modulation#Modulation_index

4. Lab 2 Assignment

#0. Write an overview of your results.

#1. Open the AM_Mod_coherent.slx file. This program has been pre-coded for you. We have a baseband (message) signal whose amplitude can be varied by the first tunable knob named "Baseband Signal: Amplitude" while the simulation is running in Simulink. The Carrier signal has a fixed amplitude of 2 but its frequency can be modified by the second knob named "Carrier signal: Frequency". The third knob allows you to view the spectrogram of different signals.

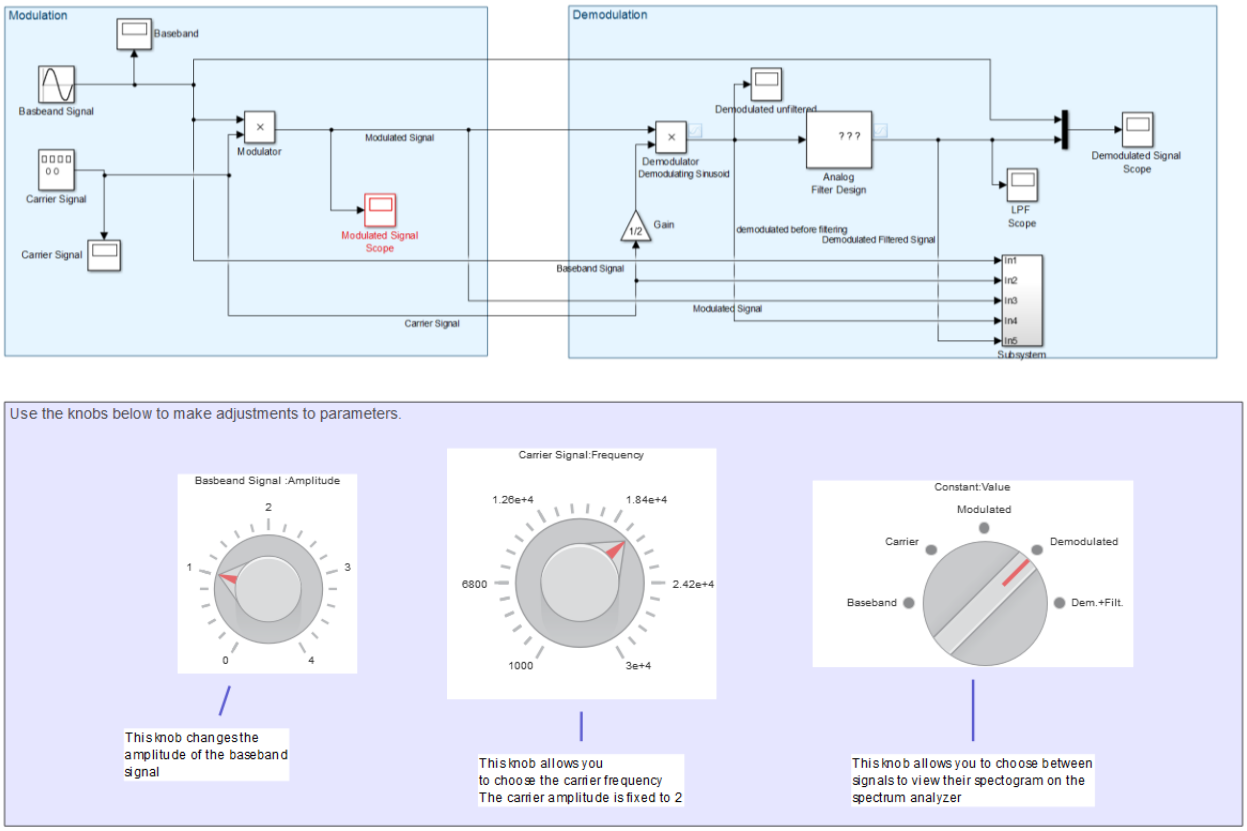
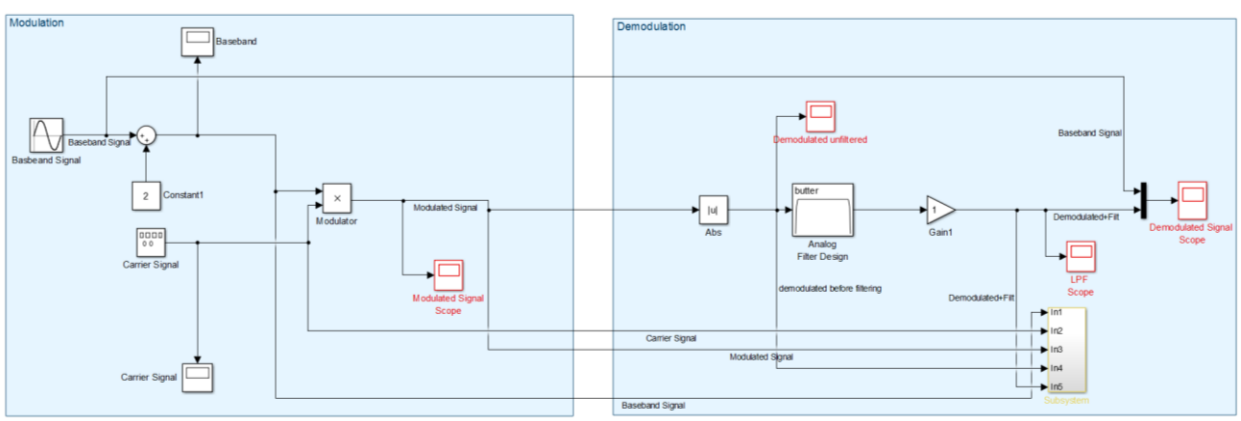


Figure 1. Modulation and Coherent Demodulation

1. In the "Analog Filter Design" block, the parameters of Butterworth filter have been assigned using variables for the sake of this exercise: "n" is the order of the filter and "cutoff" is the cutoff frequency in rad/s. By looking at the frequency of the "Baseband Signal" (2000Hz), decide on a cutoff value and order for the filter. Assign values to "n" and "cutoff" in the MATLAB command window so that the variables are in the workspace. Run the simulation. On the "Demodulated Signal Scope", the baseband signal and the one that is modulated and demodulated should have the same frequency and amplitude with maybe some phase shift. What is the choice of "n" and "cutoff" that gives you perfect demodulation? Provide screenshot of the "Demodulated Signal Scope" and show that the frequency and amplitude are indeed the same (phase can be different).

2. With the parameters “n” and “cutoff” chosen for the filter in 1, provide screenshots of the spectrum analyzer for all the 5 positions in knob 3. These will show you the spectrogram of the baseband, carrier, modulated signal, demodulated signal and finally the demodulated and filtered signal. If the choice in 1 was correct, the baseband and demodulated+filtered signal should look very similar. Provide side-by-side screenshots of those two to show that is the case.
3. The exercise starts off with the carrier signal at 20,000Hz. Without changing the butterworth filter from 1, vary the carrier frequency signal (knob 2 “Carrier signal: Frequency”, or enter it in the frequency parameter inside the “Carrier Signal” block) to 10,000Hz, 5,000Hz and 1,000Hz. Provide screenshots of the spectrogram from the baseband signal and the demodulated+filtered signals (change knob 3) in each case. Discuss your observations.
4. We will now see the effect of modulation index. Keep the carrier frequency at 20,000Hz. Change the baseband amplitude from 1 to 2 and then to 4 by changing the parameter inside the “Baseband Signal” block. For both cases, provide spectrograms of the baseband vs demodulated+filtered signal (change knob 3) and discuss your observations. Do they match with what was taught in Section 3 in the lab manual?

#2. Open the AM_Mod_incoherent.slx file. This program has been pre-coded to perform amplitude modulation and demodulation using the non-coherent method (envelope detection). Once again, the baseband (message) signal’s amplitude can be varied by the first tunable knob “Baseband Signal: Amplitude” while the simulation is running in Simulink. The carrier signal has a fixed amplitude of 2 but its frequency can be modified by the second knob “Carrier signal: Frequency”. The third knob allows you to view the spectrogram of different signals. The Butterworth filter in the “Analog Filter Design” block is a 10th order bandpass filter.



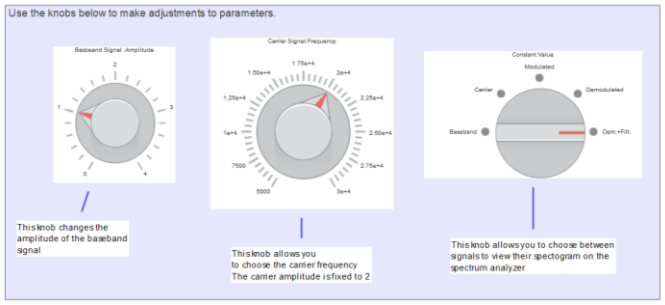


Figure 2. Modulation and Non-coherent Demodulation (Envelope Detection)

The process of non-coherent modulation requires the baseband signal to be shifted up by the amplitude of the carrier signal so as to get a completely positive signal. The constant block after the baseband signal achieves this purpose. Modulation is done by the carrier signal, and for demodulation a method called envelope detection is performed. Here, the absolute value of the received signal is taken then bandpass filtered to get the demodulated signal. In this case, however, perfect demodulation does not guarantee that the demodulated+filtered signal amplitude will match that of the baseband signal.

1. Start, by setting the Wlo value to 30*2π rad/s and Whi to 5000*2π rad/s by typing in the "Analog Filter Design" parameter box. This will set you up for correct demodulation. Take screenshots from all scopes in the block diagram (Baseband, Carrier Signal, Modulated Signal Scope, Demodulated unfiltered, LPF Scope and Demodulated Signal Scope). Explain in your own words what is happening to the baseband signal at each scope position (what we are looking for is the description of how non-coherent demodulation is happening).
2. Change the Wlo parameter to 1*2π rad/s. What happens to the spectrogram of the demodulated+filtered signal (provide screenshot)? What frequencies do you think are present in there and why?
3. We will now see the effect of modulation index in non-coherent demodulation. Switch Wlo back to 30*2π rad/s. The carrier signal has an amplitude of 2. The first knob allows us to change the amplitude of the baseband signal. All previous exercises had the baseband amplitude at 1. What is the modulation index at this point?
4. Now change the amplitude of the baseband signal so that the modulation index is 1. Is there any difference in the demodulated+filtered signal (show both spectrogram and signal vs time plots).
5. Finally change amplitude of the baseband signal so that the modulation index is 2. Provide screenshots of the spectrogram and any of the scope outputs you think is pertinent, describe what is happening.

#3. Open the AM_Mod_coherent_tada_inc.slx file. This program has been pre-coded to perform amplitude modulation and demodulation using the coherent method. However, two blocks are missing and the "Dem Gain" block is configured for the wrong gain. When missing blocks are filled in and the correct parameters are chosen for each block, the tada sound should be heard in both the direct path and the modulated/demodulated path. The two paths are chosen via the "Manual Switch" during simulation.

Fill in the missing blocks and run the simulation with correct parameters. Switch the positions of the “manual switch” to make sure that you do in fact hear similar sound on both channels. Provide your choices of parameters for the two missing blocks and the corrected gain for the “Dem Gain” block. Justify your choices. Provide a screenshot of the “Demodulated Signal Scope” to show the signals from the two paths.